MAPLE & MAIN · CUSTOMER ACQUISITION

# More new customers from the same outreach budget

A propensity scorecard that finds 60% more customers at a fixed budget, and the one number to measure before spending more.



Synthetic data. No client, employer or proprietary information.

EXECUTIVE SUMMARY

# Rank prospects by propensity now. Test what outreach causes before raising the budget.

## 1,294

more new customers a year when outreach reaches 40% of prospects chosen by the scorecard instead of at random. Worth $554k in first-year margin.

## 9x

The top tenth of prospects converts at 24%, the bottom tenth at 3%. A simple scorecard captures all of the signal the data holds.

## 51%

of bottom-tenth conversions must be caused by outreach for those contacts to pay back. Nobody has measured that share.

**Recommendation:** use the scorecard to order the outreach list today, buy lead lists filtered on profile instead of dropping sources, and run a 10% holdout for two quarters to learn how many conversions outreach really causes.

APPROACH

# From a prospect list to a budget decision

Maple & Main collects about 52,000 prospects a year from five lead sources. About one in ten becomes a loyalty member within 45 days.

### 1. Who converts?

Three ranking methods compared on a later period: a scorecard, gradient boosting, and a look-alike index.

SQL · Python · SAS

### 2. What is a customer worth?

First-year margin measured for 7,440 prospects who became customers, including those who never ordered.

Linked to 2.3M orders

### 3. What does a ranked list buy?

Customers found at every budget level, against a random list.

Out-of-time test, 49,805 prospects

### 4. What is still unknown?

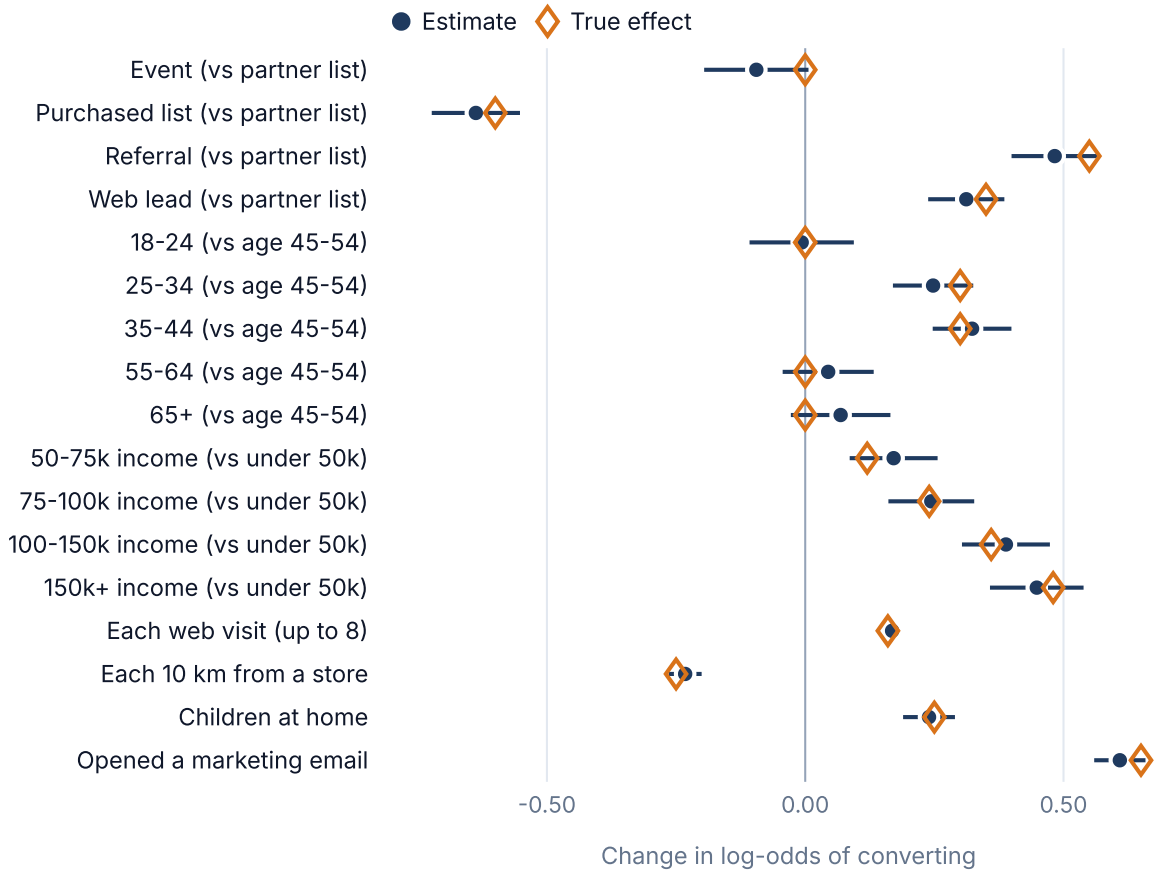
Whether outreach causes the conversion. The break-even is worked out for each score band, with the test to measure it.

Sensitivity and test design

| DATA USED | CHECKS BEFORE MODELLING | HOW RESULTS WERE VALIDATED |
|---|---|---|
| 150,000 prospects with lead source, profile and web behaviour, linked to the customer and order files | 10,643 prospects removed because their 45-day conversion window was not fully observed; eight integrity checks passed | Trained on 2024 to September 2025, tested on the following year, and compared with the conversion rule built into the data |

WHAT DRIVES CONVERSION

# Email engagement, lead source and distance to a store matter most




Effect on the log-odds of converting, with 95% intervals. Diamonds show the true effect built into the synthetic data.

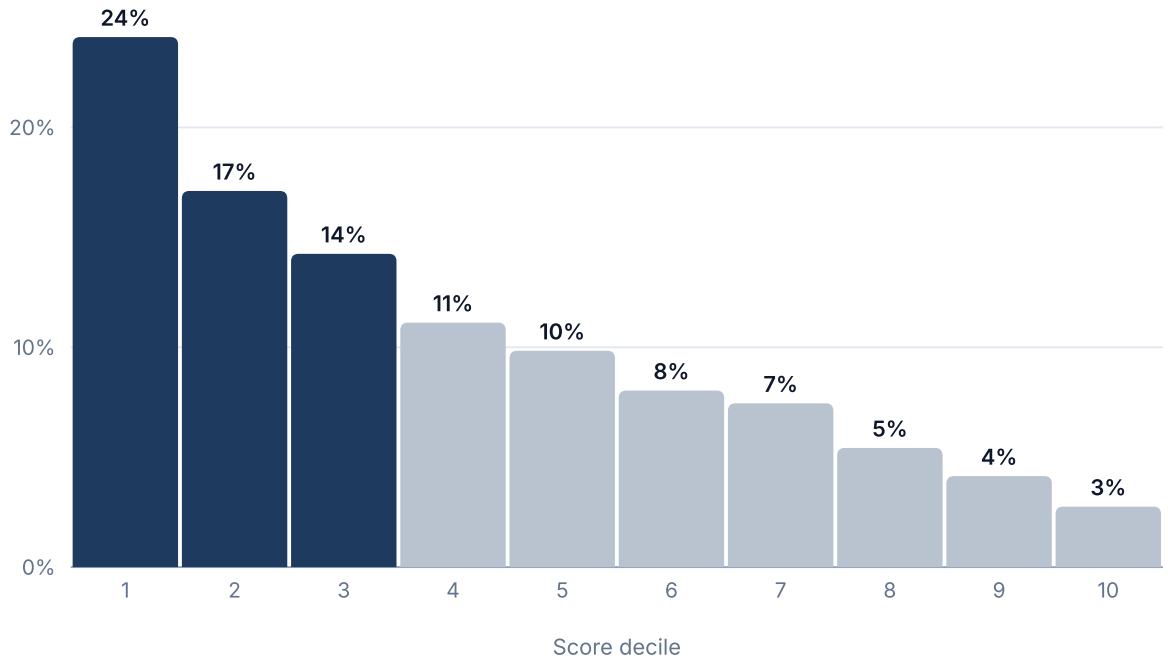
- A prospect who has opened a marketing email has about **1.8 times** the odds of converting.
- Referrals and web leads convert best. Purchased lists convert worst.
- Every 10 km from a store cuts the odds by about **21%**.
- Ages 25 to 44, households with children and higher incomes all convert more.

> All 17 estimated effects fall within their margin of error of the true value. The scorecard has learned the real mechanism, not noise.

MODEL PERFORMANCE

# An AUC of 0.68 looks modest. It is also the most this data can give.

**Conversion rate by score decile, prospects from October 2025 to September 2026**




Decile 1 is the highest score. Average conversion is 10.4%.

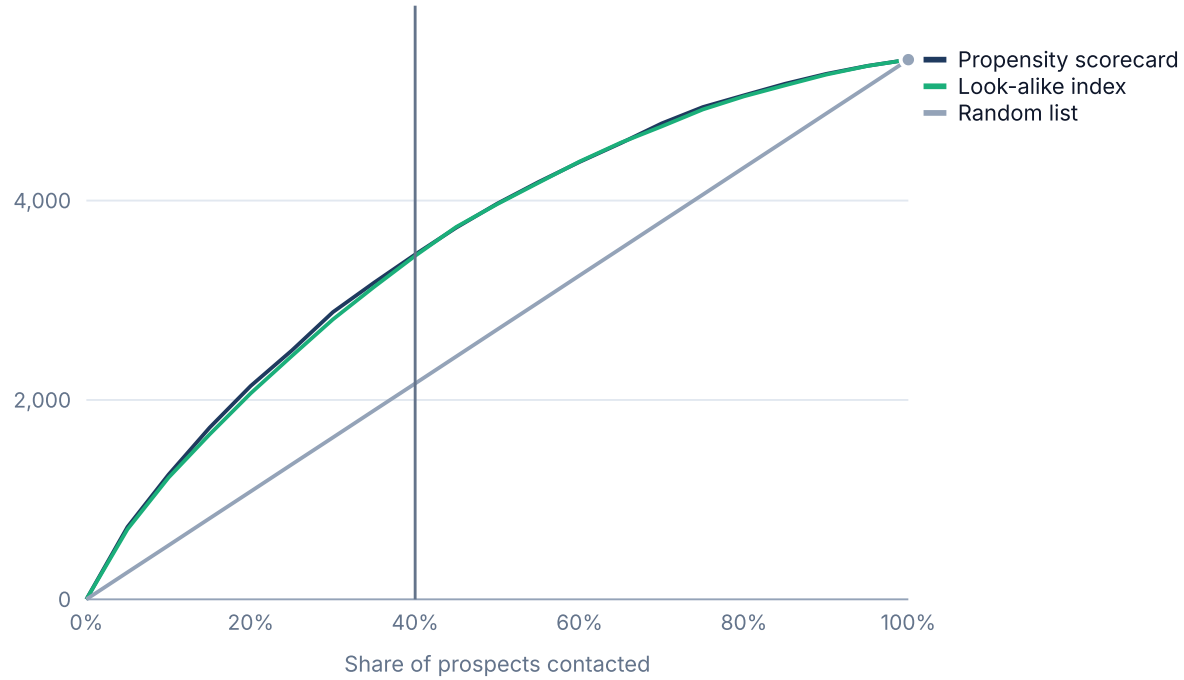
| RANKING METHOD | AUC | SHARE OF ACHIEVABLE SIGNAL |
|---|---|---|
| Web visits only | 0.60 | 52% |
| Look-alike index | 0.68 | 97% |
| Gradient boosting | 0.68 | 97% |
| **Propensity scorecard** | **0.68** | **100%** |
| Ceiling: the true conversion rule | 0.69 | 100% |

- Conversion is partly chance. Even a perfect model would score 0.69 here.
- The more complex model does slightly worse than the scorecard. More modelling will not help; new data might.
- Conversion has been rising, so scores read low: 9.4% predicted against 10.4% actual. The ranking is unaffected; the baseline needs a quarterly refresh.

BUSINESS CASE

# At the same budget, the ranked list finds 1,294 more customers a year

**New customers found, by share of prospects contacted**




Prospects from the test year. The look-alike index sits almost exactly on the scorecard line.

| PROSPECTS CONTACTED | RANDOM LIST | SCORECARD | EXTRA FIRST-YEAR MARGIN |
|---|---|---|---|
| 20% (10,388) | 1,082 | 2,141 | $453k |
| **40% (20,776)** | **2,165** | **3,459** | **$554k** |
| 60% (31,164) | 3,249 | 4,387 | $488k |
| 80% (41,552) | 4,331 | 5,056 | $310k |

Customers a year. Margin uses the measured $428 first-year margin per new customer.

> At 40% reach, outreach cost per new customer falls from $58 to $36. The look-alike index does as well as the scorecard, so a ranked list is possible even without records of who did not convert.

LEAD SOURCES

# Profile matters more than source: a good purchased-list name beats a weak referral

### Conversion rate by lead source and score band

| LEAD SOURCE | BOTTOM THIRD | MIDDLE THIRD | TOP THIRD |
|---|---|---|---|
| Event | 5.0% | 8.4% | 17.3% |
| Partner list | 5.0% | 8.4% | 14.5% |
| Purchased list | 4.0% | 9.1% | 8.7% |
| Referral | 7.0% | 8.6% | 18.0% |
| Web lead | 6.2% | 10.2% | 18.7% |

Score band is the scorecard third across all prospects.

### Cost to acquire a customer, by source

| LEAD SOURCE | CONVERTS | COST PER LEAD | COST PER CUSTOMER |
|---|---|---|---|
| Web lead | 15.5% | $8 | $90 |
| Partner list | 8.8% | $5 | $126 |
| Purchased list | 5.1% | $2 | $158 |
| Referral | 14.1% | $20 | $184 |
| Event | 8.6% | $15 | $245 |

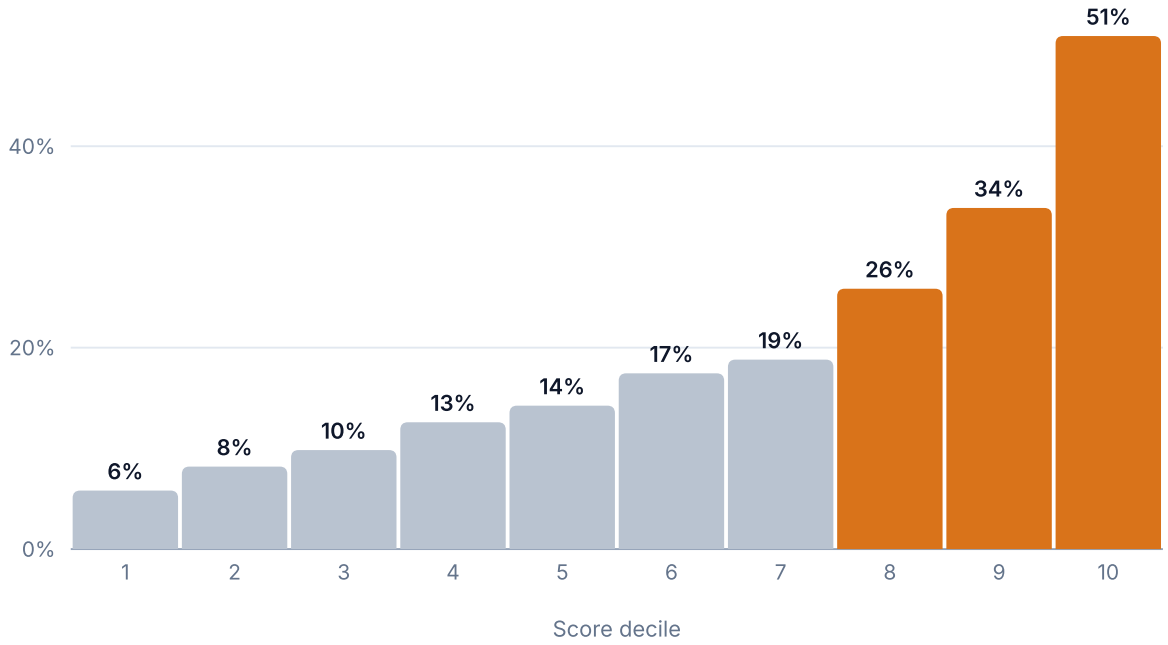
Cost per lead is an assumption, not data. Cost per customer adds $6.00 of outreach per prospect. Change both in the simulator.

> Cutting purchased lists would lose good prospects. Buying them filtered on income, age, children and distance keeps the low price and lifts conversion.

THE NUMBER NOBODY HAS MEASURED

# Every contact pays back only if outreach causes the conversion

**Share of conversions outreach must cause for a contact to break even**




By score decile. Break-even is the $6.00 contact cost divided by conversion rate times $428 first-year margin.

Some prospects would have joined anyway. The data records who converted, not who converted *because* they were contacted. There has never been a holdout.

| IF OUTREACH CAUSES | NET A YEAR, CONTACT ALL | DECILES LOSING MONEY |
|---|---|---|
| 100% of conversions | $2007k | 0 |
| 75% of conversions | $1427k | 0 |
| 50% of conversions | $848k | 1 |
| 25% of conversions | $268k | 3 |

> The top deciles are safe under almost any assumption. The case for contacting the bottom third rests on a number that has not been measured.

TEST DESIGN

# A 10% holdout for two quarters settles it

**Smallest effect the test can detect, in conversion points**

| HOLDOUT SHARE | ONE QUARTER | TWO QUARTERS | ONE YEAR |
|---|---|---|---|
| 5% | 3.4 (33%) | 2.4 (23%) | 1.7 (17%) |
| **10%** | **2.5 (24%)** | **1.8 (17%)** | **1.3 (12%)** |
| 20% | 1.9 (18%) | 1.3 (13%) | 0.9 (9%) |

80% power, 5% two-sided significance, at 52,000 prospects a year and a 10.4% conversion rate. In brackets: the smallest share of conversions caused by outreach that the test would detect.

- **Design:** hold back a random 10% of prospects from outreach for 26 weeks. Compare 45-day conversion.
- **What it answers:** whether outreach causes at least 17% of conversions, and how that varies by score band.
- **Cost:** about 2,600 prospects go uncontacted. If outreach is fully effective, that forgoes roughly $116k of first-year margin, less the contact cost saved.
- **Decision rule:** expand outreach to every score band whose measured incremental value exceeds the contact cost.

RECOMMENDATIONS

# Four actions

1 **Order the outreach list by the scorecard**

It is a 17-term formula that runs in SQL. Refresh the baseline every quarter because conversion is drifting up.

**$554k a year**

at 40% reach

2 **Buy lead lists on profile, not just price**

Filter purchased and partner lists on income, age, children and distance to a store before buying.

**Depends on list pricing**

3 **Run the 10% holdout for two quarters**

Until then, treat any plan to extend outreach to low-scoring prospects as unproven.

**Decides the budget**

4 **Stop tuning the model; add new signals**

The scorecard already captures all the signal in these inputs. Product interest, basket intent or postal-code demographics are the way to a better ranking.

**Raises the ceiling**

APPENDIX

# Method, assumptions and limits

**Population.** Prospects first contacted from 1 January 2024 to 15 September 2026. Later prospects are excluded because conversion can take 45 days.

**Target.** Became a loyalty member within 45 days of first contact.

**Validation.** Trained on 89,552 prospects to September 2025, tested on 49,805 from the following year.

**Scorecard.** Unregularised logistic regression on lead source, age band, income band, web visits (capped at 8), distance, children and email engagement.

**Look-alike index.** For each trait, the share of converters with it divided by the share of all prospects with it; a prospect's score is the product of its indices.

**Customer value.** Mean first-year margin of converted prospects with a full year of orders. 9% never ordered and are included at zero.

**Assumed, not measured.** Outreach cost of $6.00 per prospect, and cost per lead by source. Both can be changed in the simulator.

**Not known.** How many conversions outreach causes. All "extra customer" figures compare ranked and random lists of contacted prospects and assume outreach is what converts them.

**Limits.** First-year margin understates lifetime value. A single average value is used for every converted prospect; in this data the score does not predict how valuable the customer turns out to be, so that simplification costs little here but should be re-checked on real data.

**Synthetic data.** Maple & Main is fictional. The data was generated with a known conversion rule, which is why the model can be compared with the truth. Figures illustrate the method, not any real company.

**Code.** SQL, Python and SAS, with a model card and QC log, are in the project repository.